1 Introduction

During this sprint, after the kick-off meeting, we sat down and read the available material regarding the platform, regulations, and proposed documentation. After that, we had a meeting where we decided in which direction the project is supposed to go, and how we wanted it to get there. Due to the lack of a hardware platform (car), we spent our time setting up our workspace, repairing and initiating simulations, collecting the initial dataset, writing a line detection algorithm and developing a middleware to allow more modular development.

2 Planned activities

During this sprint, we had a plan to accomplish the following tasks:

- Familiarizing ourselves with the platform, regulations, and documentation.
- Setting up and starting the simulator
- Deploying code on the Nucleo and RPi platforms
- Implementing and deploying line detection

3 Status of planned activities

- **Familiarizing ourselves with the platform, regulations, and documentation**
  We spent a significant amount of time reading the documentation, regulations and reviewing the code, as well as attending the discussion meetings in an effort to fully understand what is expected of us. We tried to separate critical tasks from less important ones to avoid losing time unnecessarily. At this moment we understand the basic concepts of the platform, but we are prepared to invest additional time when we start to work on specific tasks.

- **Deploying code on the Nucleo and RPi platforms**
  Since the delivery of the car was delayed, we have been unable to complete this task. In the meantime, we were told we would be able to pick up the platform on the 17th and we intend to get on this task as soon as we can.

- **Simulation starting**
  After successfully overcoming minor challenges, we started the simulation and can drive the car in it. Although we attempted to follow the tutorial in the provided documentation, we encountered an issue where the car in the simulation did not receive the messages sent from the keyboard. We spent several days resolving this problem, and eventually, we identified the need to export an additional variable enabling Gazebo to use controllers.
  We plan to use this simulation to gather training data for a YOLO model.

- **Implementing and deploying line detection**
  We implemented a number of line detection algorithms in an effort to find the most optimal for our needs. In the video, you can see it in action. As stated before, we weren't able to deploy it due to the platform not arriving yet.

4 General status of the project

The car can be driven using the keyboard in the simulation, we have a decent line detection algorithm and are working on training a YOLO model, but it should all be integrated into one project. We have also started developing a middleware component which will serve as a communications medium between the brain and the simulation. The idea is to have a separate module which translates simulation info to useful data for the brain and vice versa. That way there is less strain being put on the brain which doesn't need to process any information and only needs to react to what the middleware is telling it. We will try our best to make this idea fully come to life in the next couple of sprints.

5 Upcoming activities

In the next sprint we want to do:

- Traffic sign detection and line detection in simulation
- Deploy code on Nucleo and RPi, and set everything up
- Collect images from simulation and additional model training
- Collect images from the car camera
- Making physical a environment
- Completely disassemble and reassemble the car for maintenance
- Further development of middleware